

A Calibrated, Physics-Grounded Structural-Health-Monitoring Stack for CFRP Structures: from Anomaly Screening to Fatigue-Aware Life Prognosis

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Draft — June 11, 2026

Abstract

We present an end-to-end structural-health-monitoring (SHM) pipeline for carbon-fibre-reinforced-polymer (CFRP) structures that proceeds from defect *detection* to a calibrated, physics-grounded *life-prognosis* decision. On full-field thermoelastic stress data of a perforated CFRP interstage, we (i) benchmark nine anomaly detectors and show that a simple per-node statistic attains AUROC ≈ 0.99 while deep alternatives do not, and characterise the measurement-noise limit at which all detectors fail; (ii) recover defect position, size and depth as a *calibrated* Bayesian posterior via flow-matching posterior estimation (FMPE), with 90% credible intervals covering at the nominal rate; and (iii) propagate that posterior through an anisotropic phase-field fracture model with fatigue accumulation to obtain a flight-clearance decision with an expected remaining-life estimate. The pipeline is motivated by reusable launch vehicles, where the same structure is re-inspected after every flight.

1 Introduction

2 Data and problem setting

3 Stage 0 — anomaly screening

Methods. Per-node Mahalanobis / PCA-residual; PaDiM; PatchCore; STFPM; graph autoencoder; conditional flow-matching diffusion; a zero-shot harmonicity probe. Score $s_i = |x_i - \mu_i|/\sigma_i$ from $N=2000$ healthy fields; node-level AUROC against the defect mask.

Result. Per-node statistics dominate (Table 1, Fig. 2). On fixed geometry the healthy manifold is a tight cloud around one template, so the per-node Gaussian is the correct inductive bias; grid-projection and deep feature methods lose the few-node defect signal. All methods collapse by additive noise $\sigma=0.1$ (defect amplitude $< 0.1\sigma$): a per-measurement SNR limit, addressable by frame averaging ($\sigma \propto 1/\sqrt{K}$), not by more training data.

4 Stage 1 — classification and type identifiability

Is the type signal real, or a temporal/temperature confound? On the long-term guided-wave set, damage states were introduced in different years, so a naive type classifier can alias damage type with season/temperature. We test this directly. Restricting to seven damage tags whose temperature supports overlap (pairwise overlap 0.93), we compare four conditions: temperature-only (acc = 0.16, near chance), waveform-only (0.91), waveform+temperature (0.92), and waveform with temperature *partialled out* of every feature (0.91). The net type discrimination beyond temperature is $\Delta\text{acc} = 0.758$ (95% CI [0.735, 0.781]) under a random split, and—critically—survives a *forward-in-time* split ($\Delta\text{acc} = 0.184$,

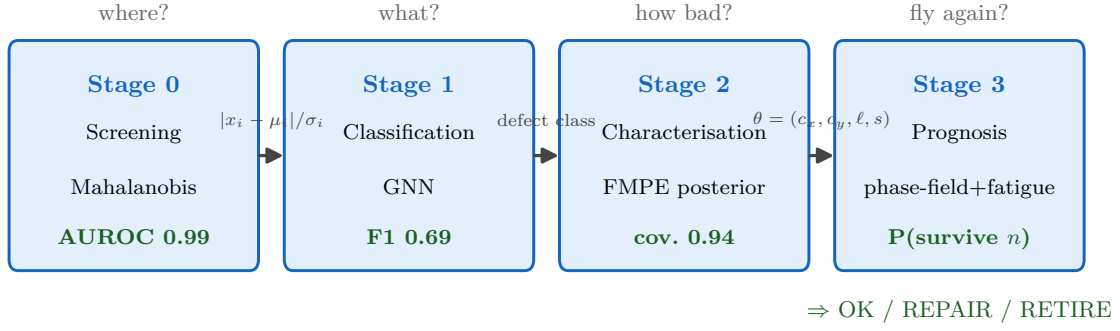


Figure 1: Stage 0–3 pipeline. Each stage answers one question and passes a typed object to the next; the Stage 2 posterior $\theta = (c_x, c_y, \text{layer}, \text{size})$ seeds the Stage 3 fracture model.

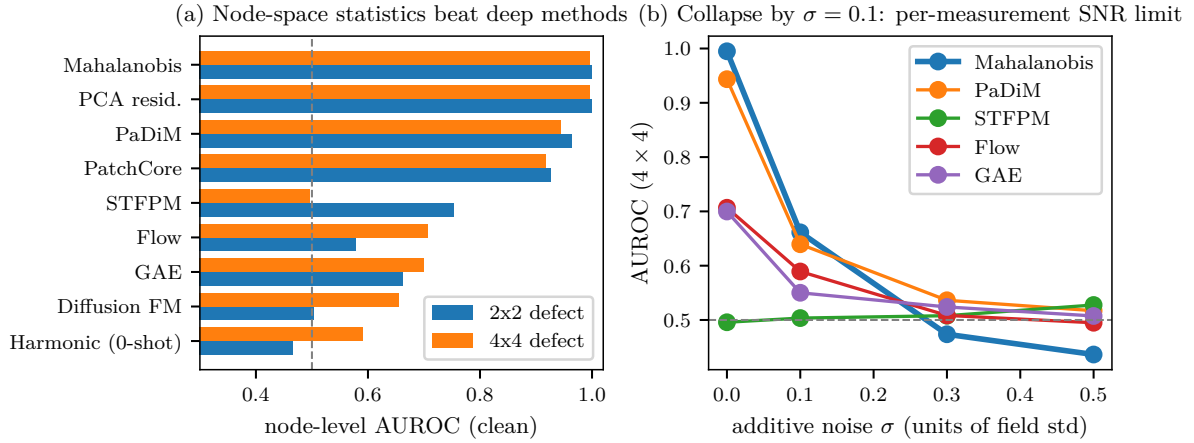


Figure 2: (a) Clean node-level AUROC by method. (b) Collapse with measurement noise; the defect signal sits below 0.1σ .

95% CI [0.166, 0.203]). Both lower bounds exceed zero: the type posterior reflects a genuine damage signal, not the environmental confound. (An earlier single-month binary probe, where temperature-only matched the majority baseline, was an artefact of that restricted setting.)

5 Stage 2 — calibrated defect characterisation (FMPE)

6 Stage 3 — phase-field prognosis and flight clearance

7 Discussion

8 Conclusion

Reproducibility. All stages are implemented and tested (19/19 unit tests) in two public-on-request repositories; figures are generated by the scripts under `paper_figs/`.

Table 1: Anomaly detectors, clean node-level AUROC (2x2 / 4x4 defects).

method	2x2	4x4
per-node Mahalanobis	0.999	0.995
PCA residual	0.999	0.995
PaDiM	0.964	0.944
PatchCore	0.927	0.918
STFPM	0.754	0.496
graph autoencoder	0.663	0.700
flow (RealNVP)	0.578	0.707
diffusion (FM)	0.503	0.663
harmonicity (0-shot)	0.465	0.590

Table 2: FMPE posterior on held-out test fields.

parameter	RMSE	R^2	90% coverage
position c_x	0.055	0.78	0.944
position c_y	0.021	0.83	0.940
size (3-class)	—	0.73	0.894
layer (depth)	2.98	0.63	0.870

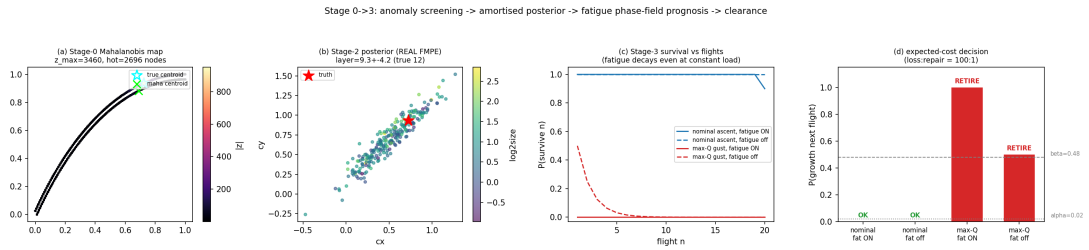


Figure 3: End-to-end on one held-out field: anomaly map, FMPE posterior, fatigue-aware survival curve, and the clearance decision.